

Bismark A. Agyei

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<https://github.com/kofimuad>

Profile

I am a versatile professional with a background in metallurgical engineering and front-end software development, complemented by experience in materials science, process control, audio and video editing. Passionate about continuous learning, I am expanding my skills in SolidWorks, welding, and renewable energy. With a track record in teaching, research, and creative writing, I thrive in multidisciplinary roles and aim to contribute innovative solutions in engineering, technology, and energy.

Technical Skills

Programming Languages: Python, C, R, HTML, CSS.

Engineering Tools: Visual Studio Code, GitHub.

Laboratory Techniques: Atomic Absorption Spectrometry (AAS), X-Ray Fluorescence (XRF).

Software Proficiency: Aspen Plus, COMSOL Multiphysics, MATLAB, LIMS, SolidWorks, Microsoft Office Suite, Git.

Professional and Research Experience

Student Energy Fellowship Program | *Student Energy, Vancouver, Canada (Remote)* **March– Sept. 2024.**

- Conducted research on the feasibility and implementation of small modular reactors (SMRs) in Ghana.
- Gained expertise in project management and strategic planning within renewable energy initiatives.
- Researched and analyzed data to evaluate Ghana's energy landscape and identify opportunities for nuclear integration.

Teaching Assistant | *Materials Engineering Dept. - KNUST, Obuasi, Ghana.* **Nov. 2022 - Aug. 2023.**

- Conducted laboratory analyses using AAS and XRF, ensuring precision and adherence to safety protocols.
- Managed maintenance and calibration of laboratory equipment to meet established standards.
- Assisted in instructing courses on phase transformations, ferrous and non-ferrous metallurgy, numerical methods, and thermodynamics.
- Supported the development of laboratory procedures and training materials for students.

Research Project | *Recovery of Gallium from Waste LED Bulbs – KNUST.* **Jan. – Sept. 2022.**

- Investigated methods for gallium recovery using techniques like crushing, grinding, and chemical extraction.
- Optimized lixiviants for effective dissolution and recovery of gallium from waste materials.
- Analyzed the purity and concentration of recovered gallium using AAS and XRF techniques.

- Product Developer (Volunteer)** | *Ishara Recycling, Kumasi – Ghana.* Jan. 2022.
- Conducted user research to gather information on customer needs.
 - Developed a prototype software solution based on gathered insights.
- Marketing and Sales Intern** | *Slamm Technologies, Accra, Ghana.* Sept. 2021.
- Supported the implementation of marketing strategies and promotional campaigns.
 - Conducted market research to identify growth opportunities and enhance brand awareness.
 - Prepared detailed reports and presentations on marketing strategies using Microsoft Office tools.
- Audio Editor** | *Gold Coast Report (GCR), Accra – Ghana (Remote).* Jan. 2020 – July 2023.
- Edited and enhanced audio recordings for clarity, quality, and consistency across multiple podcast episodes.
 - Utilized audio editing software (e.g., Audacity, Adobe Audition) to remove background noise, adjust levels, and ensure professional sound quality.
 - Designed and implemented sound effects, transitions, and music to improve audience engagement.
 - Ensured timely delivery of polished audio files to meet production schedules.
 - Maintained an organized library of audio assets for future use and reference.
 - Provided feedback and suggestions to improve podcast audio and overall production quality.

Education

Kwame Nkrumah University of Science and Technology Aug. 2018 – Sept. 2022.
B.Sc. Metallurgical Engineering
Relevant Coursework: Physical Metallurgy, Extractive Metallurgy, Thermodynamics and Kinetics, Process Control and Simulation, Materials Characterization, Laboratory Techniques, Research Methodologies.

References

- **Dr. Emmanuel Tsiwah** - Lecturer, KNUST, Obuasi Campus
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 - **Email:** ea.tsiwah@knust.edu.gh.
- **Dr. Martinson A. Nartey** - Lecturer, KNUST, Kumasi and Obuasi Campuses.
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